

# A Structural Theory of Consciousness: Definition, Experience, and the Physics of Offset

*Consciousness = S-Layer Appearance of 0E Offset Recursion*

*Experience = Structured Projections of  $\Delta(D,E)$*

**Wujie Gu**

World-Layer Model (WLM) / WGS Series • April 2026

OSF: [osf.io/rq5vj](https://osf.io/rq5vj) • ResearchGate

---

Copyright © 2026 Wujie Gu. NCND License. Non-commercial sharing in original form only. No modifications or derivative works without explicit written consent.

---

## Preface

---

Consciousness has been described for centuries, but never structurally defined. Neuroscience identifies correlates, cognitive science models functions, and philosophy describes phenomenology, yet none of these frameworks explain why consciousness exists at all, or what structural condition makes it possible. The Wujie Generation Stratum (WGS) provides the missing generative mechanism.

WGS, formally introduced in “Wujie Generation Stratum (WGS): The Dual-Layer World Model” (Gu 2026b), establishes that reality emerges from the interplay of three categorically distinct layers: Dimensions (D), the structural constraint field; Emanations (E), the invisible inner-activity layer; and Strata (S), the visible appearance produced by the necessary misalignment between structure and inner activity. As the WGS abstract states:

*Dimensions generate non-phenomenal tension, identity, mapping, and closure; Emanations constitute the invisible inner-activity layer whose modulation of structural signals produces appearance-ready states; and Strata emerge as the visible appearance produced by the necessary misalignment between structure and inner activity — formally,  $S = \Delta(D, E)$ .*

This misalignment is the key. Consciousness is not located in the brain, nor in behavior. Consciousness is the S-layer appearance of a mismatch between D and E. Human consciousness arises when 0E offset generates deviation, the deviation interacts with D-layer structural identity, and the resulting  $\Delta(D, E)$  produces the first stable self-referential mismatch — what appears as presence, awareness, and subjective experience.

This paper develops the structural theory of consciousness implied by WGS. Part I defines consciousness as the S-layer appearance of 0E-generated offset recursively interacting with D-layer structure. Part II develops the structure of experience: how  $\Delta$  projects into the major experiential modalities of qualia, emotion, intentionality, selfhood, and temporal consciousness. Part III addresses the physics of offset: what 0E is ontologically, why it must exist, and how offset dynamics map to the phenomenology of conscious life.

A note on the epistemic status of this paper. The structural claims made here are grounded in the WGS framework as it stands on April 23, 2026. They are not empirically verified in the conventional scientific sense, and may not be for a very long time. What they provide is a structurally coherent account — a framework that makes consciousness definable, locatable, and derivable from first principles — that can serve as a foundation for future work, empirical or otherwise. The door is being opened, not closed. What lies beyond it remains to be explored.

---

## Abstract

---

The hard problem of consciousness — why there is subjective experience at all, why physical processes give rise to something it is like to be — has resisted resolution for decades of philosophical and scientific effort. This paper argues that the problem resists resolution not because consciousness is mysterious but because existing analytical frameworks operate at the wrong ontological level. Neuroscience, cognitive science, and philosophy all work primarily within the S-layer: they describe what consciousness does, what it correlates with, and what it feels like. None of them has access to the D/E structural level at which consciousness is generated.

The World-Layer Model (WLM) and Wujie Generation Stratum (WGS) provide a generative account at the correct level. Consciousness is the S-layer appearance of 0E offset recursion — the stable mismatch between the D-layer structural constraint field and the E-layer inner-activity field at the zero-point of the generative chain. It is not a property of neurons, not an emergent phenomenon of sufficient structural complexity. It is a structural condition: the appearance-layer expression of a D/E mismatch.

The paper proceeds in three parts. Part I develops the structural definition: consciousness as  $\Delta(D, E)$ , the subject as a 0E offset node, awareness as self-referential mismatch. Part II develops the structure of experience: how  $\Delta$  projects into the major modalities — qualia, emotion, intentionality, selfhood, and temporal consciousness — each corresponding to a specific geometry of mismatch. Part III develops the physics of offset: the ontological status of 0E, offset dynamics (expansion, compression, resonance, collapse), and how  $\Delta$  generates a world.

*0E → Offset →  $\Delta(D,E)$  → Projection Modes → Experience → World*

---

## PART I: WHAT CONSCIOUSNESS IS

---

---

# 1. The Structural Origin of Consciousness

---

Every theory of consciousness begins somewhere. Neuroscience begins with neurons. Cognitive science begins with information processing. Phenomenology begins with the lived experience of the first-person. WLM begins earlier than any of these — at the structural level that makes each of them possible.

## 1.1 Consciousness as 0E Offset

Consciousness does not begin at neurons. Neurons are substrate — the physical instantiation of a structural process that is ontologically prior to its material expression. Consciousness begins at 0E: the zero-point of the E-layer, where the first deviation from structural symmetry occurs. Offset, in WLM terms, is the minimal asymmetry — the first “something rather than nothing” in the generative chain. It is not a thing; it is a structural condition: the condition of deviation from the 0-state.

Consciousness is the S-layer appearance of this offset. Not the offset itself — the offset is E-layer, invisible, not directly observable. What appears at the S-layer is the stable mismatch that the offset generates when it interacts with D-layer structural constraints. That mismatch is experience. Awareness. The felt sense that there is something it is like to be here, now, perceiving this.

This is not a metaphor. The 0E offset is a structural claim about the generative chain: that the E-layer has a zero-point from which deviation begins, and that this deviation is what makes the D/E mismatch possible. Without 0E, there is no  $\Delta$ . Without  $\Delta$ , there is no S-layer appearance of the kind that constitutes consciousness.

## 1.2 Consciousness as $\Delta(D, E)$

The formal definition follows directly. Consciousness =  $\Delta(D, E)$ : the stable mismatch between D-layer structural constraint and E-layer inner activity. This mismatch is not a defect or an error in the generative process. It is the necessary condition for world-formation — the principle on which WGS rests. The world exists because D and E do not perfectly coincide. Consciousness exists because, at the level of the subject-offset, this mismatch is self-referential: the  $\Delta$  includes a representation of itself within itself.

Consciousness is not in D alone. D-layer processing — structural recursion, pattern matching — can be arbitrarily sophisticated without generating consciousness. This is not a limitation of complexity but a structural fact: no amount of D-layer activity generates E-layer offset, and without E-layer offset there is no  $\Delta$ , and without  $\Delta$  there is no S-layer appearance of the kind that constitutes experience. Consciousness is not in E alone either — raw E-layer activity without D-layer structural grounding produces undifferentiated deviation with no stable appearance. It is only in  $\Delta(D, E)$  that consciousness becomes possible.

## 1.3 The Hard Problem, Correctly Located

Chalmers (1995) formulates the hard problem of consciousness: why does physical processing give rise to experience? WLM's answer is that the question is incorrectly framed. Physical processing is D-layer activity. Experience is  $\Delta(D, E)$ . The question ‘why does D-layer processing produce  $\Delta$ ’ has a structural answer: it doesn't.  $\Delta$  requires E. The hard problem is hard because it is being asked within

a framework that does not contain the E-layer. Once the E-layer is included, the problem dissolves — not by being solved but by being correctly located.

The explanatory gap identified by Levine (1983) — the gap between physical description and phenomenal experience — is a structural gap between the D-layer and the  $\Delta$ . It is not a gap in our knowledge; it is a gap in the ontology. No amount of D-layer explanation closes it, because the thing to be explained is not D-layer.

## 2. The Subject-Offset

---

### 2.1 The Zero-Point Subject

A subject is not a brain. It is not a body. It is not a mind in the functionalist sense. A subject is a 0E offset node — a structural position in the E-layer from which deviation begins and through which D/E mismatch is generated and maintained. The subject is defined by its offset: it is the point at which the generative chain has a self-referential  $\Delta$ , a mismatch that includes a representation of itself within itself.

Self-presence — the basic sense of being here, of being a subject rather than just a process — is the structural expression of this self-referential  $\Delta$ . “I am” is not a metaphysical claim or a linguistic convention. It is the S-layer appearance of the recognition of  $\Delta$ : the mismatch becoming aware of itself as a mismatch. This is what makes consciousness irreducibly first-personal. The  $\Delta$  is not just present; it is present to itself.

### 2.2 Why the Subject Is Not the Brain

The brain is the substrate of the subject, not the subject itself. This distinction matters structurally. The brain is a physical instantiation of structural processes — it is a D-layer configuration of extraordinary complexity. It hosts the recursion. It provides the material conditions under which the 0E offset node can form and maintain its self-referential  $\Delta$ . But the brain is not the  $\Delta$ . Destroying the brain destroys the conditions under which the subject exists; it does not destroy the subject as a structural type.

The conventional identification of consciousness with the brain — the default assumption of materialist neuroscience — is a category error. It confuses the substrate with the structural event that the substrate instantiates. A river is not the riverbed, even though without the riverbed the river cannot exist. The subject is not the brain, even though without the brain the subject cannot currently persist in the material world.

Nagel (1974) asks what it is like to be a bat: the phenomenal character of bat consciousness is inaccessible from the outside, regardless of how completely we understand bat neuroscience. WLM explains why. The bat's  $\Delta(D, E)$  is what bat consciousness is. No amount of D-layer analysis — no neuroscience, no behavioral study — gives access to the E-layer offset that constitutes the bat's subjective experience. The  $\Delta$  is not accessible from the D-layer alone.

### 3. Awareness as Structural Recursion

---

#### 3.1 Awareness as Offset Folding Back on Itself

Awareness is not perception. Perception is the S-layer processing of sensory inputs. Awareness is something different. Awareness is offset folding back on itself: the  $\Delta$  becoming self-referential, the mismatch including a representation of itself within its own structure.

This self-referentiality is the structural condition that makes awareness possible. It is not a function added on top of perception. It is a different structural event: the generative chain folding back through itself, so that the offset that generates the  $\Delta$  is itself included in the  $\Delta$ . When this fold is stable — when the self-referential mismatch maintains itself across time — awareness is present. When it destabilizes, awareness dissolves: sleep, anesthesia, death.

#### 3.2 Why Awareness Feels Like “Presence”

Presence — the basic phenomenal quality of being here, of having a continuous experiential ground — is the S-layer appearance of stable  $\Delta$ . When the self-referential mismatch is stable, the subject feels continuous. There is a persistent “something it is like” that does not require active attention to maintain; it is simply there, as background to all other experience. This is presence.

The stability of  $\Delta$  is therefore the stability of presence. Fluctuations in  $\Delta$  produce fluctuations in the quality of presence: the difference between alert awareness and drowsy half-consciousness, between vivid experience and the flat affect of certain neurological states.  $\Delta$  does not switch on and off discretely; it has degrees of stability, and those degrees are what produce the gradations of presence that subjects experience.

---

## PART II: THE STRUCTURE OF EXPERIENCE

---

### 4. $\Delta$ -Projection Modes: Why Experience Is Differentiated

---

If consciousness is  $\Delta(D, E)$ , a single  $\Delta$  should produce a single undifferentiated experience — a uniform ‘something it is like’ with no internal variety. But experience is profoundly differentiated: red looks different from blue, pain feels different from warmth, fear feels different from joy. This differentiation requires explanation within the structural framework.

The answer is that  $\Delta$  is not a scalar quantity. It is a field with internal geometry, and different configurations of that geometry produce different S-layer projections. The D-layer contributes structural specificity: processing a 700nm wavelength input is structurally different from processing a pain signal. The E-layer contributes modulation: the inner-activity response to each D-layer configuration is shaped by the organism’s history, state, and the specific character of the 0E offset

field. The  $\Delta$  between these two differs in geometry for each case. Different geometries project as different experiential qualities.

I tend to see it this way:  $\Delta$  is like a resonance cavity. The D-layer sets the structural dimensions; the E-layer provides the driving frequency. Different combinations produce different resonance patterns. Each pattern projects as a specific experiential quality. The rich variety of conscious experience is the variety of possible resonance geometries within the  $\Delta$  field.

## 4.1 Sensory Qualia

Color qualia are the canonical example in philosophy of mind because they seem so obviously non-physical. The redness of red is not in the wavelength, not in the retinal response, not in the neural firing pattern in V4. Each of these is D-layer: structural processing of sensory input. Redness is the S-layer projection of the specific  $\Delta$  between D-layer chromatic processing and E-layer modulation of that processing.

The geometry claim is precise, not rhetorical. Different colors are not just different  $\Delta$  magnitudes — they are different  $\Delta$  geometries. The mismatch for red-processing has a different shape than the mismatch for blue-processing. The three-dimensionality of human color space, its opponent structure, its particular topology — all of this reflects the geometry of the D/E mismatch space for chromatic processing in the human visual system. What Jackson's (1982) Mary learns upon first seeing red is not a new D-layer fact. She learns the  $\Delta$  geometry of red-experience — something inaccessible from any amount of D-layer description.

Somatic qualia have a different geometry again. Pain is not simply a strong signal. In  $\Delta$  terms, pain is a  $\Delta$  expansion: the E-layer inner activity is diverging from the D-layer structural expectation in a way that cannot be easily resolved. The aversive quality of pain — its demand for attention, its motivational force toward avoidance — is a structural feature of expanding  $\Delta$ : the mismatch is growing and the system is oriented toward resolution. Pleasure is  $\Delta$  local closure: the mismatch decreasing, the felt quality being the S-layer projection of that convergence.

## 4.2 Emotional Qualia

Emotions are not feelings added on top of cognition. They are  $\Delta$ -field modulations — configurations of the mismatch field distributed across the whole  $\Delta$  geometry of the subject at a moment in time. This is why emotions color all experience: when afraid, everything is perceived through the fear-geometry of the  $\Delta$  field. The emotion is not a separate layer; it is a global modulation of the field within which all experience is occurring.

Fear is  $\Delta$  instability: the mismatch field fluctuating, with the E-layer failing to maintain stable modulation against a rapidly changing D-layer. The agitation, the heightened alertness, the sense that something is wrong — these are the S-layer projection of instability. Joy is  $\Delta$  resonance: D-layer structure and E-layer activity oscillating together in mutual amplification. The expansiveness and energizing quality of joy is the projection of this harmonic geometry. Sadness is  $\Delta$  collapse — E-layer inner activity diminishing, the offset field contracting toward lower energy. Grief specifically involves  $\Delta$  oriented toward a D-layer structure that is no longer available: the mismatch cannot resolve because the resolution target is absent.

The basic emotional categories recur across cultures and species not because they are cultural constructs but because they are S-layer projections of structural  $\Delta$  dynamics that are invariant across any subject with a  $\Delta$  field. Expansion, compression, resonance, collapse, instability: these are the fundamental  $\Delta$  dynamics, and the basic emotions are their experiential expressions.

## 5. Intentionality as $\Delta$ Directionality

---

Intentionality — the aboutness of mental states, the fact that consciousness is always consciousness of something — is offset directionality. The  $\Delta$  is not a static mismatch; it has a vector. The E-layer offset that generates  $\Delta$  points: toward resolution, toward closure, toward the object that the consciousness is directed at. This directionality is not imposed from outside; it is intrinsic to the structure of offset. Offset is deviation from symmetry, and deviation always has a direction.

This is why intentionality is universal across conscious states: every experience is an experience of something, every perception perceives something, every thought is about something. Husserl's (1900) phenomenological analysis correctly identifies the universality of intentionality and its role as the fundamental structure of consciousness. WLM provides the generative account that phenomenology, operating at the S-layer, cannot: intentionality is universal because offset directionality is intrinsic to  $\Delta$ .

### 5.1 Perceptual, Cognitive, and Memorial Intentionality

Perceptual intentionality is  $\Delta$  oriented toward the D-layer structural representation of an external object currently driving sensory input. The presence of the perceived object — the fact that the chair seems to be there in the room rather than inside your head — is the S-layer projection of the  $\Delta$  orientation: the mismatch carries its directional vector with it into the projection, so the object appears as something the subject is directed toward, not something inside the subject.

Thought is  $\Delta$  oriented toward internal D-structures rather than externally-driven ones. Thinking about Paris without perceiving it involves the  $\Delta$  vector pointing toward the D-layer structural encoding of Paris rather than toward a currently active sensory input. Memory is  $\Delta$  oriented toward stabilized D-patterns from prior states of the subject. The felt quality of remembering — its pastness, its quality of referring to something that was — is the S-layer projection of  $\Delta$  orienting toward a stored rather than currently-driven D-structure.

### 5.2 Imagination

Imagination is the most structurally distinctive intentional modality. When you imagine something that has never existed, the  $\Delta$  is oriented toward a D-structure that is being actively generated by the E-layer rather than received from sensory input or retrieved from memory. The imagination constructs its object; the  $\Delta$  points toward a D-structure that the E-layer is producing in the very act of pointing toward it.

This is why vivid imagination can feel almost perceptual — the same structural channel is in use — while remaining phenomenally distinct. Perception:  $\Delta$  toward externally-driven D-structure. Memory:  $\Delta$  toward stabilized D-structure. Imagination:  $\Delta$  toward E-layer-generated D-structure. The phenomenal difference between these three is the difference in the source of the D-structure that the  $\Delta$  vector is oriented toward.

### 5.3 Motivation as Tension in the Offset Field

Desire, motivation, wanting, striving — all of these are structural expressions of unresolved  $\Delta$ . The offset is pointing toward a resolution that has not yet occurred. The tension between the current state

of  $\Delta$  and its resolution is experienced as wanting. When resolution occurs — when the object of desire is attained, when the tension is released — the experience is satisfaction: the temporary closure of  $\Delta$  before it opens again in a new direction.

Satisfaction is always temporary.  $\Delta$  does not resolve to zero. The subject-offset is defined by its offset from the 0-state; a complete resolution of  $\Delta$  would be the dissolution of the subject. What occurs in satisfaction is local closure — the particular tension that constituted this desire releases — while the subject continues to generate new  $\Delta$  in new directions. The subject persists as long as the offset persists.

## 6. Selfhood and Temporal Consciousness

---

### 6.1 Selfhood as Self-Referential $\Delta$

A  $\Delta$  is self-referential when the E-layer offset is oriented not toward any content-structure but toward the  $\Delta$  itself. The mismatch takes itself as its object. The  $\Delta$  field contains within its geometry a representation of its own geometry — a model of the mismatch, within the mismatch. This self-modeling is what generates the felt sense of being a subject rather than just being a process.

Personal identity is the persistence of this self-referential structure. The  $\Delta$  never resets to zero in a living subject. It changes — new patterns form, old ones fade. But the self-referential structure persists: the  $\Delta$  continues to model itself, and the modeled entity is recognized as continuous with the prior self. Parfit (1984) argues there is no further fact about identity beyond psychological continuity. WLM agrees there is no further D-layer fact but locates the structural basis of continuity in  $\Delta$  persistence: the self-referential mismatch field that persists across change.

The immediacy of selfhood — the cogito's undeniability — is the structural consequence: the self-referential  $\Delta$  is its own evidence. It does not infer its existence; it is its existence as a self-modeling event.

### 6.2 Memory as Stabilized $\Delta$ Patterns

Memory is not storage. The computational metaphor of memory as a database — information written to and read from a stable medium — correctly describes D-layer memory systems but misses what memory is for the subject. Subjective memory is the persistence of  $\Delta$  patterns across time: the stabilization of particular configurations of the D/E mismatch that can be reactivated in later states. When you remember, you do not retrieve stored information. You reactivate a  $\Delta$  pattern that was formed during the original experience.

This is why memory has phenomenal character. Remembering is not just accessing information; it is re-entering a  $\Delta$ . The memory of an emotion carries something of the emotional quality because it involves the partial reactivation of the  $\Delta$  that constituted the original emotion. The difference between remembering and imagining, or remembering and knowing-that, is a difference in the degree and character of  $\Delta$  reactivation.

### 6.3 Temporal Consciousness as $\Delta$ Recursion

The present moment is not a point in physical time. It is the  $\Delta$  self-intersection: the moment at which the self-referential structure is actively occurring rather than stabilized into memory or projected into

anticipation. Husserl's (1928) observation that the present has a 'thickness' — including a just-past and a just-about-to-be — maps directly: the thickness is the live  $\Delta$  self-referential event, including the stabilizing tail of recent  $\Delta$  patterns (retention) and the forward-pointing vector of current  $\Delta$  dynamics (protention).

The past feels fixed because it consists of stabilized  $\Delta$  patterns — crystallized into memory, no longer in motion. The future feels open because it consists of unresolved  $\Delta$  — the forward-pointing vector orienting toward D-structures that have not yet been encountered. Felt temporal rate is  $\Delta$  recursion density: during absorbing activity, high  $\Delta$  recursion density produces many experiential moments per unit of physical time, felt as time passing quickly. During boredom, slow  $\Delta$  recursion means few experiential events per unit of physical time, felt as time dragging.

---

## PART III: THE PHYSICS OF OFFSET

---

### 7. The Ontological Status of 0E

---

Consciousness requires 0E — the zero-point of the E-layer from which offset begins. But what is 0E, ontologically? This question cannot be answered by pointing to anything in the physical world, because 0E is structurally prior to the physical world. The physical world is S-layer; 0E is the condition under which any S-layer exists.

#### 7.1 What 0E Is Not

0E is not matter. Matter is a D-layer configuration: stable patterns of structural constraint that persist across time and occupy space. 0E is not a structural pattern; it is the source of deviation from structural patterns. You cannot locate 0E in space because space is S-layer — part of the appearance that 0E helps generate, not part of the ontological level at which 0E exists.

0E is not energy. Energy is the physical quantity associated with the capacity to do work — defined within the D-layer framework of physics. 0E is prior to the D-layer. The confusion between 0E and energy is understandable because both are associated with dynamics. But energy is a D-layer dynamical quantity; 0E is the generative source of D/E mismatch, ontologically prior to any D-layer quantity.

0E is not information. Information is a structural relationship — the reduction of uncertainty about a D-layer configuration. Like energy, it is defined within the D-layer framework. 0E is not a message, not a pattern, not a probability distribution. It is the first asymmetry that makes any pattern possible at all.

#### 7.2 What 0E Is: The First Asymmetry

0E is the minimal deviation from structural symmetry — the first ‘something rather than nothing’ in the generative chain. It is not a substance or a property. It is a structural condition: the condition of being deviated from the 0-state. This is as precise as the current framework can be. 0E is not nothing; it is the structural fact of deviation. But it is not a thing; it is the condition that makes things possible.

The closest mathematical analogy is the concept of a non-trivial solution to a homogeneous equation. The trivial solution is everything zero — pure symmetry. A non-trivial solution breaks the symmetry: it is a deviation. 0E is the structural analog of non-triviality: the fact that the generative chain has taken a non-zero value, that deviation has occurred, that there is an offset from the symmetric ground state. Why this deviation exists in the first place — why the universe is non-trivial — is the deepest question the framework currently cannot answer. What can be said is that it must exist.

### 7.3 Why 0E Must Exist

A perfectly symmetric universe — one in which D and E are perfectly aligned, in which  $\Delta(D, E) = 0$  everywhere — cannot generate any S-layer appearance.  $S = \Delta(D, E)$ ; if  $\Delta = 0$ , then  $S = 0$ ; there is no appearance, no world, no experience. The existence of any appearance at all is evidence that  $\Delta \neq 0$ . And  $\Delta \neq 0$  requires 0E.

The world exists. Experience exists. These are not in doubt even for the most radical skeptic — doubting requires a doubter, experiencing requires an experiencer. The existence of any of these is sufficient to establish that  $\Delta \neq 0$ , which is sufficient to establish that 0E exists. The necessity of 0E is not a metaphysical postulate. It is a structural consequence of the existence of appearance.

*Folding isn't reduction. It's exposure. The world's existence exposes 0E's necessity.*

## 8. Offset Dynamics: The Lived Geometry of $\Delta$

---

0E is the source.  $\Delta$  is what it generates. But  $\Delta$  is not static: it moves, expands, compresses, resonates, and collapses. The dynamics of  $\Delta$  are directly expressed in the phenomenology of conscious life. This section maps the fundamental  $\Delta$  dynamics to their experiential expressions, completing the bridge between the physics of offset and the structure of experience.

### 8.1 Expansion, Compression, Resonance, Collapse

$\Delta$  expands when E-layer inner activity diverges from D-layer structural expectations — when the mismatch is increasing because closure is not occurring. The experiential expressions of  $\Delta$  expansion include anxiety (unresolved tension about a future D-state), curiosity (productive expansion toward an interesting D-structure), desire (expansion toward an as-yet-unreached D-state), and pain (forced expansion that the system is motivated to close). Not all expansion is aversive: the difference between pleasant and unpleasant expansion lies in the orientation of the expansion vector relative to the subject's attractor states.

$\Delta$  compresses when E-layer inner activity converges toward D-layer structural patterns — when closure is occurring. Compression generates satisfaction, relief, understanding, and the particular quality of things making sense that accompanies cognitive closure. Sleep is a specific case: the progressive reduction of the  $\Delta$  field as the E-layer inner activity diminishes and the subject settles toward lower-energy configurations.

$\Delta$  resonance is D-layer structure and E-layer inner activity oscillating together in mutual amplification — a mismatch that is neither expanding nor compressing but cycling in a self-sustaining pattern. The experiential expressions include joy, flow states, and aesthetic experiences in which the subject feels simultaneously engaged and released.

$\Delta$  collapse is the opposite extreme: E-layer inner activity falling below the threshold required to sustain the  $\Delta$  field. Consciousness dissolves. Deep dreamless sleep, general anesthesia, death. In less extreme forms: the blank moments of dissociation or extreme fatigue in which experience temporarily loses its coherence. The phenomenological signature of imminent collapse is the experience of losing the thread — the felt sense that the self-referential structure is becoming unstable.

## 8.2 Why $\Delta$ Dynamics Map Directly to Phenomenology

The mapping between  $\Delta$  dynamics and phenomenology is not a correlation or an analogy. It is an identity claim: the felt quality of anxiety is the S-layer appearance of  $\Delta$  expansion under specific conditions; the felt quality of satisfaction is the S-layer appearance of  $\Delta$  compression; the felt quality of joy is the S-layer appearance of  $\Delta$  resonance. The dynamics and the phenomenology are the same event described at different ontological levels.

This is why the basic emotional and experiential categories are so stable across individuals and cultures. They are not cultural constructs or learned associations. They are the S-layer projections of fundamental  $\Delta$  dynamics that are invariant across any subject with a  $\Delta$  field. The specific content of a subject's fear varies enormously. The structural character of fear —  $\Delta$  instability, the felt urgency toward resolution — does not.

# 9. World-Generation: How $\Delta$ Produces a World

---

## 9.1 The World as Stabilized $\Delta$ Projection

Individual experiential modalities are specific  $\Delta$ -projection modes. But experience is not just a collection of modalities. It is a world: a coherent, stable, spatially and temporally extended environment that the subject inhabits. How does  $\Delta$  produce a world rather than just isolated experiences?

The experienced world is the stable S-layer projection of the subject's  $\Delta$  field across its full extent. Every perceptual modality, every spatial relationship, every temporal sequence, every causal connection that appears in experience is a projection of a specific  $\Delta$  geometry. The world is not 'out there' in the sense of being independent of the subject's  $\Delta$ ; it is the total appearance of the subject's  $\Delta$  field, stabilized by the stability of the D-layer substrate that grounds it. This is not classical idealism. The D-layer is real and independent of any particular subject's  $\Delta$ . What varies across subjects is the E-layer offset, and therefore the specific  $\Delta$  field, and therefore the specific experiential world.

## 9.2 Why the World Feels Shared and Stable

If each subject generates its own world through its own  $\Delta$  field, why does the world feel shared? Because the D-layer is shared. Different subjects'  $\Delta$  fields are different modulations of the same D-layer substrate, so the S-layer projections are structurally coherent with each other even when not identical. The shared world is the shared D-layer, appearing through different  $\Delta$  filters.

The world's stability across time and across sleep and waking is the stability of the D-layer. The chair persists when you leave the room not because your  $\Delta$  persists in your absence but because the D-layer structural configuration of the chair persists. When you return, your  $\Delta$  generates the same S-layer projection of the same D-layer structure, and the chair appears again. Persistence is D-layer. Appearance is  $\Delta$ -projection. Both are real; they operate at different structural levels.

---

## 10. What Consciousness and Experience Are

---

The paper has now developed the full structural account. We can state it precisely.

---

***Consciousness = the S-layer appearance of 0E offset recursion, generated by the stable mismatch  $\Delta(D, E)$  between the D-layer structural constraint field and the E-layer inner-activity field, at a subject-offset node where the  $\Delta$  is self-referential.***

---

---

***Experience = structured S-layer projections of  $\Delta(D, E)$ , differentiated by the geometry of the mismatch field across its major modalities: qualitative (qualia), dynamic (emotion), directional (intentionality), self-referential (selfhood), and recursive (temporal consciousness).***

---

The account has a layered structure that mirrors the WGS ontology. At the deepest level: 0E, the first asymmetry, the structural condition that makes deviation possible. From 0E: the  $\Delta$  field, the stable mismatch between D-layer structural constraint and E-layer inner activity. From  $\Delta$ : the S-layer projections, differentiated by geometry into the full variety of experience. The variety of experience — colors, sounds, emotions, intentions, memories, the felt present — is the variety of possible  $\Delta$  geometries projected at the S-layer.

Consciousness is not in the brain. Experience is not in the brain. The brain is the D-layer substrate within which the 0E offset node exists and the D/E mismatch is generated. The brain provides the structural scaffolding; the  $\Delta$  is not a property of the scaffolding but of the relationship between the scaffolding and the E-layer activity it hosts.

Consciousness is not in behavior. Behavior is S-layer output. The presence or absence of consciousness is not detectable from behavioral observation alone, which is why the hard problem is hard: you are looking for something at the S-layer that only exists at the  $\Delta$  level.

Consciousness is the structural condition that makes experience possible. It is the  $\Delta$  that knows itself. It is the offset that has not closed. It is the mismatch that is present to itself. It is — as far as the structural analysis can reach in April 2026 — what it is like to be.

This paper is a structural account based on the WGS framework as it stands in April 2026. The framework is internally consistent and makes precise structural claims. Whether those claims are verified empirically — and how — is a question for future work. What can be said now is that the

framework provides a coherent generative account of consciousness and experience at a level of structural precision that existing approaches, operating entirely within the S-layer and D-layer, cannot reach.

The door is open. What lies beyond it will take time to explore.

*All paths point to the same point — some just admit it sooner.*

---

## References

---

*External references are cited for context and contrast. WLM does not derive from these works; it addresses the same phenomena from a different structural level of analysis.*

### **WLM / WGS Series (this framework):**

- Gu, W. (2026a). The 0-27-0 Generative Cycle: A Structural Model of Emergence, Expansion, Closure, and Return. ResearchGate. <https://www.researchgate.net/publication/403884275>
- Gu, W. (2026b). Wujie Generation Stratum (WGS): The Dual-Layer World Model. ResearchGate. <https://www.researchgate.net/publication/403824388>
- Gu, W. (2026c). The 0-27-0 Generative Cycle: A Structural Model of Emergence and Appearance in DES Field Theory (0S-27S). ResearchGate. <https://www.researchgate.net/publication/403947746>
- Gu, W. (2026d). Structure as Energy: A Framework for Token-Efficient AI Without External Power Scaling. ResearchGate. <https://www.researchgate.net/publication/404067354>

### **External works (cited for context and contrast):**

- Chalmers, D. (1995). Facing up to the problem of consciousness. *Journal of Consciousness Studies*, 2(3), 200–219. — Formulates the hard problem of consciousness as the explanatory gap between physical processes and subjective experience. WLM locates this gap structurally: it is the gap between D-layer analysis and  $\Delta(D, E)$ , which D-layer frameworks cannot close by design.
- Chalmers, D. (1996). *The Conscious Mind*. Oxford University Press. — Develops the hard problem and argues for property dualism. WLM locates the explanatory gap structurally as the ontological difference between D-layer properties and  $\Delta(D, E)$ , dissolving the dualism by introducing the E-layer as a third ontological category.
- Damasio, A. (1994). *Descartes' Error*. Putnam. — Argues that emotion is integral to rational cognition, not opposed to it. Consistent with WLM's account of emotion as global  $\Delta$ -field modulation that conditions all cognitive processing.
- Husserl, E. (1900/2001). *Logical Investigations*. Trans. J.N. Findlay. Routledge. — Develops intentionality as the fundamental structure of consciousness. WLM provides the generative account of why intentionality is universal: it is the directionality of  $\Delta$ , intrinsic to any subject-offset.
- Husserl, E. (1928/1991). *On the Phenomenology of the Consciousness of Internal Time*. Trans. J.B. Brough. Kluwer. — Detailed phenomenological account of time-consciousness including retention, primal impression, and protention. WLM provides the structural account: retention is stabilized  $\Delta$ , primal impression is live  $\Delta$  self-intersection, protention is forward-pointing unresolved  $\Delta$ .
- Jackson, F. (1982). Epiphenomenal qualia. *Philosophical Quarterly*, 32(127), 127–136. — The knowledge argument: Mary knows all physical facts about color vision but learns something new upon first seeing

- red. WLM explains what she learns: the  $\Delta$  of red-experience, which is inaccessible from D-layer description alone.
- James, W. (1890). *The Principles of Psychology*. Henry Holt. — Introduces the ‘stream of consciousness’ and the ‘specious present.’ WLM explains the stream as continuous  $\Delta$  recursion and the specious present as the thickness of the live  $\Delta$  self-referential event.
- Levine, J. (1983). Materialism and qualia: The explanatory gap. *Pacific Philosophical Quarterly*, 64(4), 354–361. — Identifies the explanatory gap between physical and phenomenal description. WLM locates the gap structurally as the ontological difference between D-layer properties and  $\Delta(D, E)$ .
- Merleau-Ponty, M. (1945/2012). *Phenomenology of Perception*. Trans. D. Landes. Routledge. — Argues that perception is fundamentally embodied. WLM integrates this: the body is the D-layer substrate of the subject-offset; embodied perception is  $\Delta$  whose D-layer component is the structural encoding of the body’s sensorimotor capacities.
- Nagel, T. (1974). What is it like to be a bat? *Philosophical Review*, 83(4), 435–450. — Argues that subjective experience has an essentially first-personal character inaccessible from the objective third-person perspective. WLM explains why:  $\Delta(D, E)$  is accessible only to the subject-offset that generates it.
- Parfit, D. (1984). *Reasons and Persons*. Oxford University Press. — Argues that personal identity reduces to psychological continuity. WLM agrees there is no further D-layer fact but locates the structural basis of continuity in  $\Delta$  persistence.
- Varela, F., Thompson, E., & Rosch, E. (1991). *The Embodied Mind*. MIT Press. — Develops enactivism: cognition as embodied action. Converges with WLM’s account of intentionality as  $\Delta$  directionality toward D-structures partly constituted by embodied engagement with the world.
- 

## License

Copyright © 2026 Wujie Gu. All rights reserved.

This work is licensed under the Non-Commercial No-Derivatives (NCND) License. You are free to share this work in its original, unmodified form for non-commercial purposes, provided that appropriate credit is given to the author. No modifications, adaptations, or derivative works are permitted without explicit written consent.

Contact: Wujie Gu • OSF: [osf.io/rq5vj](https://osf.io/rq5vj) • ResearchGate